

AquaSense – Phase 2

Enhanced Smart Automatic Water Dispenser

Phase: Phase 2 — Enhanced Smart Dispenser

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This report describes proposed Phase 2 improvements. The features and architecture in this document are planned development targets and are not presented as completed hardware results.

1. Introduction

AquaSense Phase 1 established a basic touchless water dispenser using an Arduino Uno, an HC-SR04 ultrasonic sensor, a 16×2 I2C LCD, and dual-servo control. The Phase 1 work covered electronic design, embedded programming, system architecture, mechanical integration planning, and development of a 3D-printable enclosure.

Phase 2 is required because the basic prototype requires improvements in reliability, usability, interaction, sensing, power support, mechanical strength, and maintenance. The next stage therefore defines the Enhanced Smart Dispenser: a more dependable and user-friendly system with better feedback, water-level awareness, adjustable operation, improved mechanical transmission, organized wiring, and a serviceable enclosure.

The Phase 2 proposal remains separate from later IoT and commercial features. ESP32 connectivity, cloud services, remote monitoring, analytics, food-grade construction, UV sterilization, and commercial compliance are reserved for future phases.

2. Phase 2 Objectives

The main objective of Phase 2 is to transform the basic automatic dispenser into a more reliable, user-friendly, and maintainable smart dispenser.

Reliability + Usability + Maintainability → Enhanced Smart Dispenser

Reliability

Improve sensing awareness, dispensing control, mechanical linkage, power support, enclosure strength, and internal wiring so that the system can operate more consistently.

Usability

Provide clear LED and LCD feedback, low-water warnings, adjustable dispensing duration, and a compact product form that is easier to understand and use.

Maintainability

Use modular construction, improved cable management, accessible internal placement, and a robust enclosure so that inspection, repair, and upgrades are easier.

3. Proposed Phase 2 Features

3.1 Water Level Monitoring

Water-level monitoring will allow the dispenser to determine whether sufficient water is available inside the container. A suitable level-sensing method may be selected according to the container geometry, required accuracy, electrical safety, and ease of maintenance. The controller can use the level information to identify normal, low, or unavailable water conditions.

When the detected level falls below the defined threshold, the system can prevent unexpected dispensing, display a warning, and request refilling. This reduces unsuccessful operation and improves user awareness. The sensing method, threshold, and accuracy must be finalized and validated during Phase 2 testing.

3.2 Low Water Warning

A low-water warning will inform the user when the container requires refilling. The warning can be shown on the LCD and supported by a dedicated LED state or another clear indication. The controller should prioritize the warning over a normal dispensing command when the water level is insufficient.

3.3 LED Status Indicators

Three dedicated status states are proposed to make system conditions visible without relying only on the LCD.

LED state	Meaning
Power	The system is powered.

LED state	Meaning
Ready	The system is ready to detect a hand and dispense water.
Low Water	The water level is below the defined operating threshold and refilling is required.

3.4 Adjustable Dispensing Duration

A fixed dispensing duration may not be suitable for every user, container, or use case. Phase 2 will investigate adjustable dispensing time through user settings, buttons, a control interface, or a software-defined configuration. A shorter duration can reduce water consumption, while a longer duration can provide a larger quantity when required.

The selected duration should be recorded and compared with the actual volume dispensed. The relationship between time, servo position, tap flow, and water volume must be measured rather than assumed.

3.5 Improved Mechanical Linkage

Phase 1 uses a servo-driven linkage concept to operate the tap. During physical integration, poor alignment, excessive friction, limited travel, backlash, or high mechanical load could reduce reliability. Phase 2 will refine the linkage geometry, servo mounting, pivot locations, connection points, and movement clearance.

The intended transition is:

Phase 1: Servo → Basic linkage → Tap

Phase 2: Servo → Improved linkage → Better mechanical control → Tap

The improved arrangement should provide smoother servo-to-tap motion, reduced mechanical stress, improved repeatability, and easier adjustment or replacement of parts.

3.6 Rechargeable Battery Support

Rechargeable battery support can make the dispenser more portable and reduce dependence on disposable batteries. Phase 2 will evaluate a suitable battery, charging circuit, voltage regulation, protection method, and safe enclosure placement. The system should prevent unsafe operation during charging or low-battery conditions where necessary.

Battery runtime, charging time, current consumption, and protection behavior should be measured during validation. The final power architecture must provide stable voltage to the controller, sensor, LCD, and servo system.

3.7 Compact and Robust Enclosure

The Phase 1 CAD model provides the starting point for a more refined enclosure. Phase 2 will identify weaknesses such as insufficient clearance, difficult access, weak mounting features, exposed wiring, poor balance, or challenging assembly. The enclosure can then be redesigned with stronger fastening, improved component placement, compact internal arrangement, better service access, and a cleaner appearance.

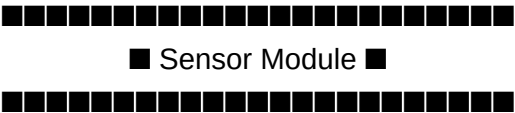
Phase 1 CAD → Identify weaknesses → Redesign enclosure → Compact Phase 2 enclosure

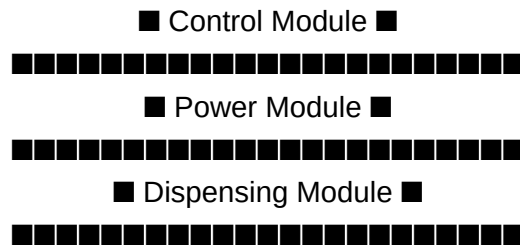
3.8 Cable Management

Improved cable management will reduce loose wires, accidental disconnections, interference with moving parts, and maintenance difficulty. Proposed methods include cable channels, clips, ties, connector labeling, strain relief, and separation of power and signal wiring where appropriate. Organized wiring will also improve the visual quality and serviceability of the product.

3.9 Modular Architecture

A modular design will divide the dispenser into replaceable and testable functional sections. This approach can simplify assembly, troubleshooting, maintenance, and future upgrades.

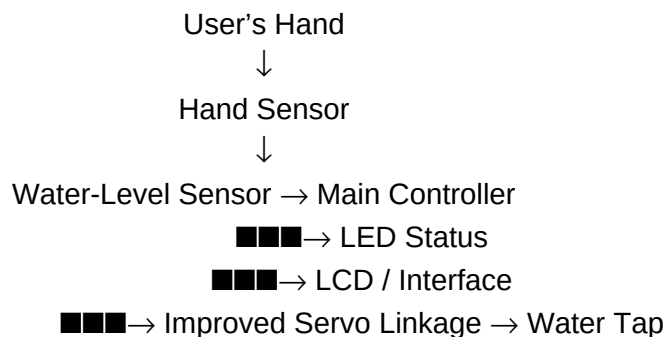




The Sensor Module may contain hand and water-level sensing. The Control Module may contain the Arduino or future controller, display interface, and logic. The Power Module may contain the battery, charger, regulation, and protection. The Dispensing Module may contain the servos, linkage, tap interface, and related mechanical parts.

4. Proposed Phase 2 System Architecture

The following architecture is conceptual and represents planned Phase 2 features, not completed hardware. It combines user detection, water-level monitoring, central control, status feedback, user interface, and improved dispensing actuation.



The controller should first evaluate water availability and user presence, then decide whether dispensing is permitted. It should provide appropriate status feedback and apply the configured dispensing duration.

5. Phase 1 vs Phase 2

Area	Phase 1	Phase 2 — Proposed
Operation	Touchless operation.	Touchless operation with enhanced monitoring and feedback.
Water monitoring	Basic or not included.	Water-level monitoring inside the container.
Low-water alert	Not included.	Proposed low-water warning.
Status indication	LCD feedback.	LCD plus Power, Ready, and Low Water LEDs.
Dispensing	Automatic operation.	Adjustable dispensing duration.
Servo mechanism	Existing linkage concept.	Improved servo-to-tap linkage.
Power	Existing prototype supply.	Rechargeable battery support.
Enclosure	Phase 1 prototype enclosure.	Compact and robust maintainable enclosure.
Wiring	Prototype wiring.	Improved cable management.
Architecture	Prototype-level arrangement.	Modular architecture.

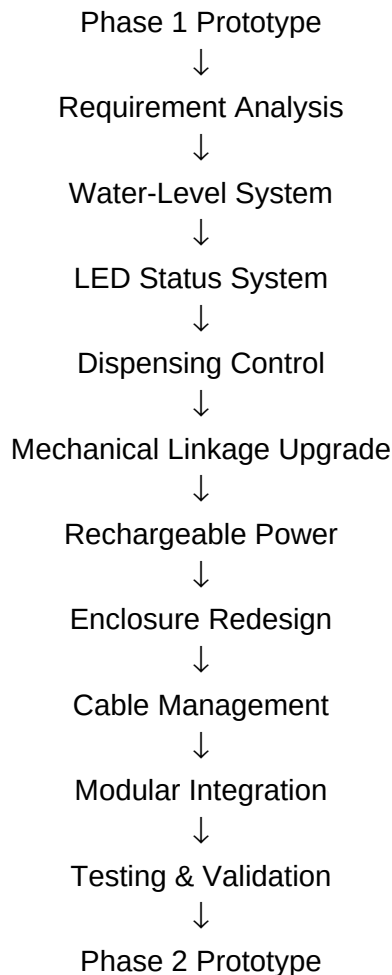
6. Expected Benefits

- Better User Experience: improved feedback, clearer system states, and more controlled dispensing.
- Reduced Maintenance: water-level awareness, modular construction, accessible components, and organized wiring.

- Improved Operational Reliability: better mechanical and electrical organization, improved linkage, and stronger enclosure design.
- Enhanced Product Appearance: compact enclosure, cleaner cable routing, and a more finished product form.

7. Phase 2 Development Plan

The proposed roadmap is:



Each stage should produce a reviewable design output, test record, or integration decision before the next stage is finalized.

8. Testing and Validation Plan

Because Phase 2 is planned, the following items are defined as validation targets rather than completed results:

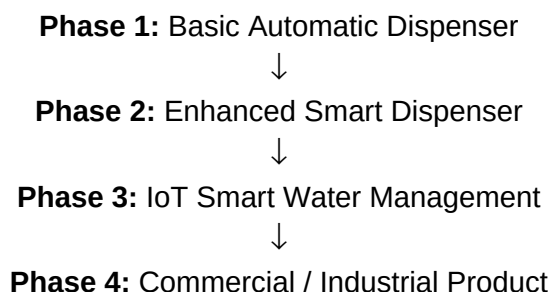
- Water-level detection accuracy and repeatability.
- Low-water warning operation at the selected threshold.
- Correct Power, Ready, and Low Water LED states.
- LCD status and distance display behavior.
- Dispensing-duration accuracy and repeatability.
- Servo linkage reliability, alignment, and mechanical stress.
- Rechargeable battery operation, charging, runtime, and protection.
- Enclosure strength, fit, access, and component clearance.
- Cable organization, connector security, and serviceability.

- Overall dispensing reliability across repeated operating cycles.

Test results should include the test condition, equipment or method, measured value, acceptance criterion, observation, and pass/fail decision. Quantitative claims should be added to the final Phase 2 report only after physical measurements are recorded.

9. Future Expansion

Phase 2 is an intermediate development stage and does not include all long-term product features.



Phase 3 — IoT Enabled Smart Water Management System

Possible Phase 3 features include an ESP32 controller, Wi-Fi connectivity, mobile integration, remote monitoring, water-consumption tracking, usage analytics, notifications, and cloud storage.

Phase 4 — Commercial and Industrial Product Version

Possible Phase 4 features include food-grade dispensing components, solenoid control, UV sterilization, automatic cleaning, improved durability, safety and compliance measures, and commercial deployment.

10. Conclusion

Phase 2 is intended to evolve AquaSense from a functional touchless prototype into a more reliable, usable, and maintainable smart dispenser. The proposed improvements focus on water-level awareness, low-water warnings, LED status feedback, adjustable dispensing, improved servo linkage, rechargeable power, compact enclosure design, cable management, and modular architecture. IoT connectivity and commercial product features remain reserved for Phases 3 and 4. The long-term vision is to develop AquaSense into a smart water-management platform that supports hygiene, water-use monitoring, reduced wastage, and sustainable consumption.